2 Trolley A moves at constant velocity towards trolley B, which is initially moving at a lower constant speed in the same direction.

Fig. 2.1 shows the trolleys at time $t = 0$.



Fig. 2.1

Table 2.1 shows data for the trolleys.

Table 2.1

trolley	mass / kg	initial speed / m s^{-1}
A	0.25	0.48
B	0.75	0.12

The two trolleys collide elastically and then separate. Resistive forces are negligible.

Fig. 2.2 shows the variation with time t of the velocity v for trolley B.

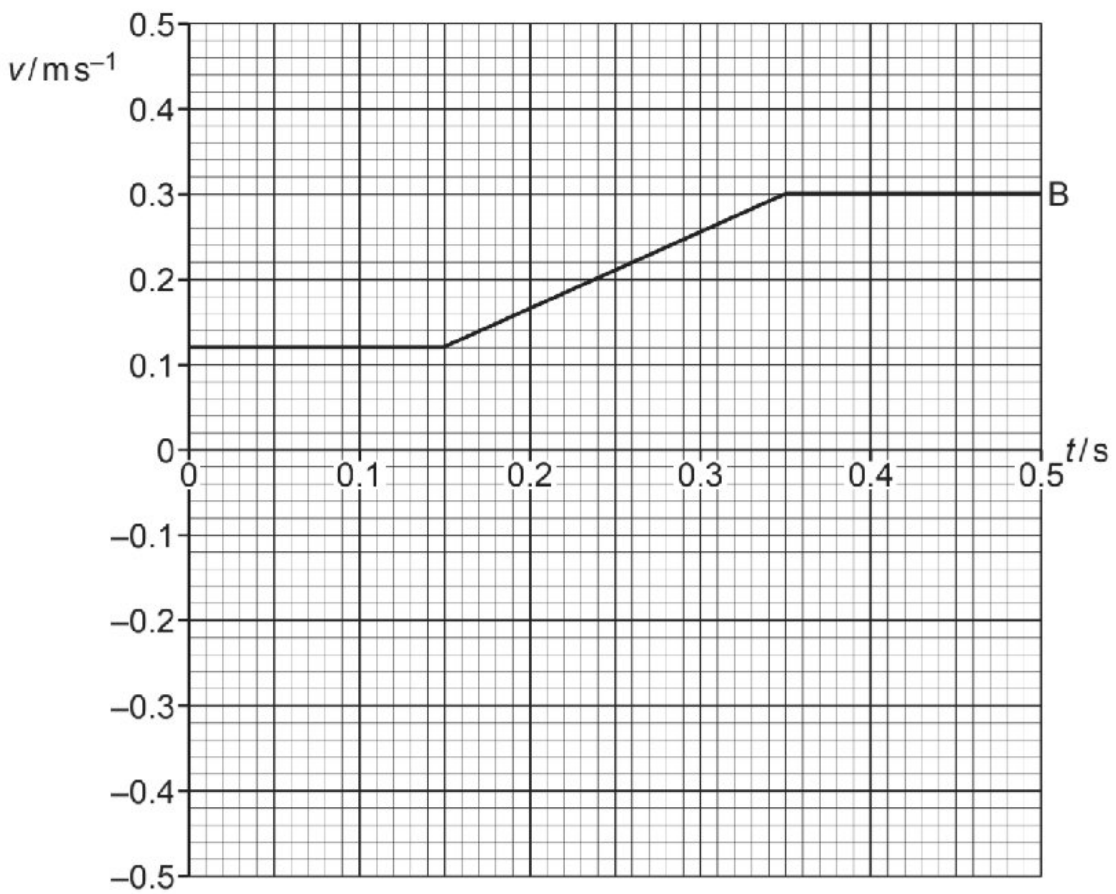


Fig. 2.2

Determine

(a) the acceleration of trolley B during the collision

acceleration of B = ... $m\ s^{-2}$ [2]

(b) the magnitude and direction of the final velocity of trolley A.

magnitude = ... $m\ s^{-1}$

direction ..... [3]

[Total: 5]